Computational Methods

Joint Inverse Problem

1. Linear solver for each tile (original formulation)
    a. Straightforward and easy to implement
    b. Serves as the benchmark/reference
    c. Practical identifiability issues
2. Bayesian inference for each tile
    a. Practical identifiability issues somewhat resolved
    b. Relatively easy to implement
    c. Dependent on priors
    d. Practical identifiability issue not fully resolved
3. Linear solver over all tiles
    a. Easy to implement (just combine everything together)
    b. Justification for combining all tiles together is not well explained
        i. Not yet formulated a way to explain how each node interacts with each other
4. Bayesian inference over all tiles
    a. Easy to implement (just combine everything together)
    b. Justification for combining all tiles together is not well explained
        i. Not yet formulated a way to explain how each node interacts with each other

Staged Inverse Problem

5. Graph Neural Network inference over all tiles
    a. Physics is explainable (Poiseuille flow correction factor; know exactly that interactions between nodes is limited to 2-4 edges)
    b. Preliminary analyses with linear solver indicate resolution to practical identifiability
        i. From synthetic graph experiments, higher distensibility values are easier to determine
    c. Correction factor abstracts some underlying physics (i.e., need to explain what contributes to the correction factor via *post-hoc* analysis)
    d. Need to search a large parameter space (not a problem on cluster)
    e. Consider DC-only vs. DC + H1 vs. DC + H1 + H2 approaches
    f. Benchmarks
        i. Vanilla Graph Convolution Network (GCN)
            1. Pure data-driven approach with no physics layer and no physics loss
        ii. Edge-local Multi-Layer Perceptron (MLP)

1. No “hops” (i.e., no message passing) to evaluate if neighbors provide useful information

iii. Linear solve vs. Bayesian solver for determining distensibility within each tile

Method	**Joint or Staged**
Linear Solver per Tile	Joint
Bayesian Solver per Tile	Joint
Linear Solver over whole mosaic	Joint
Bayesian Solver over whole mosaic	Joint
Physics Informed GNN	Staged
Vanilla GCN	Staged
Edge-local MLP	Staged

All staged methods will be further evaluated by comparing Linear and Bayesian solvers.

Experiments

Synthetic Experiments

The purpose of synthetic experiments is to evaluate how each of the methods perform under different distensibility models and noise models.

1. Choose distensibility model
    a. Constant
    b. Power Law
2. Choose noise model
3. Run simulation on whole mosaic
4. Run solvers

Real Data Experiments

1. Choose distensibility model
    a. Constant
    b. Power Law
2. Run analysis on tile or whole mosaic

3. Calculate metrics and compare findings

Features

Node Features

- Coordinates (x, y)
  - Maybe worthwhile to normalize by mosaic width and height?
- Node degree
- Radius
- Boundary type (source, sink, interior)
  - Not present in Somites27, but can be assigned
- Boundary harmonic phasors
  - Only present in Somites21, possibly best used for ablation studies
- Murray ratio
  - Only present in Somites21, probably not needed

Edge Features

- Radius
- Length
- SNR for harmonics